

PHASE 4

1. Threat Intelligence & IOC Enrichment

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Examples

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*VirusTotal

Purpose

What to Review

Example

*AbuseIPDB

Purpose

Detects

Important Field

*AlienVault OTX

Purpose

Provides

Example

*GreyNoise

Purpose

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1. Threat Intelligence & IOC Enrichment

Objective

Learn how threat intelligence helps analysts investigate alerts faster by providing context about suspicious indicators.

What is Threat Intelligence?

Threat Intelligence (TI) is information about cyber threats that helps security teams identify, understand, and respond to attacks.

Benefits

- Faster investigations
 - Better alert prioritization
 - Reduced false positives
 - Improved incident response
 - Enhanced threat detection
-

What is an IOC?

IOC (Indicator of Compromise) is evidence that suggests a system may be compromised.

Common IOC Types

Type	Example
IP Address	192.168.1.10
Domain	malicious-site.com
URL	http://evil.com/login
File Hash	SHA256 Hash
Email Address	attacker@gmail.com

2. Threat Intelligence Categories

1. Strategic Intelligence

High-level information for management.

Focus

- Threat trends
- Industry risks
- Emerging threats

2. Tactical Intelligence

Information about attacker tactics and techniques.

Examples

- MITRE ATT&CK Techniques
- Attack methods

- Adversary behavior
-

3.Operational Intelligence

Information about specific attacks or campaigns.

Examples

- Phishing campaigns
 - Ransomware operations
 - Threat actor activities
-

4.Technical Intelligence

Raw indicators used during investigations.

Examples

- IPs
 - Domains
 - Hashes
 - URLs
-

3.IOC Enrichment Process

Step 1: Identify IOC

Example:

```
IP: 185.220.101.32
```

Step 2: Query Threat Intelligence Sources

Check:

- VirusTotal
 - AbuseIPDB
 - AlienVault OTX
 - GreyNoise
-

Step 3: Analyze Results

Questions:

- Is it malicious?
 - Has it been reported before?
 - Is it linked to malware?
 - Is it part of a known campaign?
-

Step 4: Assign Verdict

Possible verdicts:

- Malicious
 - Suspicious
 - Benign
 - Unknown
-

Step 5: Map to MITRE ATT&CK

Example:

Activity	Technique
Brute Force	T1110
Phishing	T1566
PowerShell	T1059

Activity	Technique
Credential Dumping	T1003

*VirusTotal

Purpose

Checks reputation of:

- Files
- Hashes
- URLs
- Domains
- IP Addresses

What to Review

- Detection Ratio
- Community Comments
- Relationships
- Malware Family

Example

Detected by 50/70 vendors

Verdict:

Malicious

***AbuseIPDB**

Purpose

IP reputation lookup.

Detects

- Brute-force attacks
- Malware activity
- Scanning
- Botnet behavior

Important Field

Abuse Confidence Score

Score	Meaning
0-25	Low Risk
26-75	Suspicious
76-100	Malicious

***AlienVault OTX**

Purpose

Community-driven threat intelligence platform.

Provides

- Threat campaigns
- Malware families
- IOC relationships
- Threat actor information

Example

Associated with Mirai Botnet

*GreyNoise

Purpose

Distinguishes internet-wide scanners from targeted attackers.

Questions It Answers

- Is this IP attacking everyone?
- Is this attack specifically targeting us?

Possible Results

Noise

Common internet scanning activity.

Malicious

Known attacker activity.

Unknown

No sufficient information available.

*MISP

Full Form

Malware Information Sharing Platform

Purpose

Centralized threat intelligence sharing.

Features

- IOC sharing
 - Threat feeds
 - ATT&CK mapping
 - Collaboration
-

*STIX and TAXII

STIX

Structured Threat Information eXpression

Used to represent threat intelligence data.

TAXII

Trusted Automated eXchange of Intelligence Information

Used to share threat intelligence automatically.

Common Free Threat Intelligence Resources

Resource	Purpose
VirusTotal	IOC Reputation
AbuseIPDB	IP Reputation
AlienVault OTX	Threat Intelligence
GreyNoise	Scan Analysis
URLScan	URL Investigation
ThreatFox	Malware IOCs

Resource	Purpose
PhishTank	Phishing URLs
CISA KEV	Exploited Vulnerabilities

IOC Enrichment Workflow (SOC Process)

```

Alert Generated
  ↓
Extract IOC
  ↓
VirusTotal Lookup
  ↓
AbuseIPDB Lookup
  ↓
OTX Lookup
  ↓
GreyNoise Lookup
  ↓
MITRE Mapping
  ↓
Verdict
  ↓
Incident Report

```